# Jeremy Carleton

jcarleton35@gmail.com • 951-840-9209 • https://www.linkedin.com/in/jeremy-carleton/

## Education

**Texas A&M University** — College Station, TX
PhD in Computer Engineering, 4.0 GPA. — May 2028

**The University of California, Riverside** — Riverside, CA
Bachelor of Science in Electrical Engineering, 3.97 GPA. — June 2022

## Publications

**TreeHCA: Tree-Based Hindsight Credit Assignment for Multi-Turn LLM Agents** — December 2026
Debajoy Mukherjee, Jeremy Carleton, Sushil Vemuri, Srinivas Shakkottai, Dileep Kalathil
*Neurips 2026 Workshop on Long Context Foundation Models*

**Predictive Intelligent Adaptive Wireless Networks** — October 2026
Jeremy Carleton, Woo-Hyun Ko, Samin Moosavi, Avery Austin, Derek Ladd, Srinivas Shakkottai, P. R. Kumar
*2026 IEEE Military Communications Conference (Restricted)*

**MAVIS: Multi-Objective Alignment via Inference-Time Value-Guided Selection** — February 2026
Jeremy Carleton, Debajoy Mukherjee, Srinivas Shakkottai, Dileep Kalathil
*Preprint*

**CONGO: Compressive Online Gradient Optimization** — January 2025
Jeremy Carleton, Prathik Vijaykumar, Divyanshu Saxena, Dheeraj Narasimha, Srinivas Shakkottai, Aditya Akella
*The Thirteenth International Conference on Learning Representations*

**Design Implementation of A Hybrid VLP/PLC-Based Indoor Tracking System for Smart Hospitals** — February 2023
Zijin Pan, Jeremy Carleton, Sunny Zeng, Tian Lang, Gang Chen, Yehuda Kalay, Ramdas Pai, Albert Wang
*2023 IEEE International Conference on Consumer Electronics*

## Selected Coursework

**Graduate:**

- Programming With LLMs
- Wireless Communication Systems
- Deep Learning
- Advanced Optimization
- Mathematical Methods in Signal Processing
- Analysis of Algorithms
- Reinforcement Learning
- Vision Foundation Models
- Stochastic Systems
- Probability for Statistics

**Undergraduate:**

- Digital Signal Processing
- Digital Image Processing
- Modeling and Simulation of Dynamic Systems
- Software Construction
- Sensing and Actuation for Embedded Systems
- Technical Communications

## Experience

**LENS Lab at Texas A&M University** — College Station, TX
**Graduate Student Researcher** — June 2023 – Present

- Integrating formal verification into AI workflows for log analysis and planning in embodied domains
- Developing reinforcement learning algorithms tailored for long-horizon agentic tasks with sparse rewards
- Applying machine learning to resource allocation problems

**Texas A&M Engineering Experiment Station** College Station, TX
**Graduate Research Assistant** August 2023 – Present

- Applying time series prediction to tactical network management using deep learning on multi-modal data
- Done in collaboration with the U.S. Army C5ISR Center

**Bourns Inc.** Riverside, CA
**Electrical Engineering Intern** June 2021 – September 2021

- Co-led a project in the sensors & trimpots division of Bourns Inc. to build a system for mapping magnetic fields

## Selected Projects

**Pytest Agent** September 2025 – December 2025

- Designed a python application which uses collaboration between LLMs to generate test suites for python code
- Showed that over 80% code coverage could be achieved on real codebases without any human intervention

**Algorithm Design for Finding Generator Matrices of Minimum $m$-Heights** February 2025 – March 2025

- Applied evolutionary search and sparse gradient descent to efficiently discover generator matrices with small $m$-heights for real-valued analog codes

**Multi-Objective RL for Diffusion Models** September 2024 – December 2024

- Applied reinforcement learning to fine-tune a Stable Diffusion model across multiple objectives
- Demonstrated how parameter merging can be used simultaneously improve performance on multiple objectives

**Analysis of the Silver Stepsize Schedule** April 2024 – May 2024

- Analyzed why the Silver stepsize schedule improves convergence rates for unconstrained gradient descent on smooth convex functions
- Empirically studied whether the benefits of the Silver stepsize schedule extend to constrained optimization, and found that in certain cases the benefit can only be realized by adapting the schedule when projection occurs

**A Modified PyramidNet for CIFAR-10 Image Classification** March 2024 – May 2024

- Developed an image classifier based on the PyramidNet CNN architecture which achieves 95.35% accuracy on the CIFAR-10 test data

**Blockage Prediction for Wireless Communications** November 2023 – December 2023

- Combined convolutional and recurrent layers into a neural network for predicting signal loss due to blockage on a wireless channel

## Skills

**Programming Languages:** Python · C/C++ · Lua

**AI Frameworks:** Ray Distributed Training · HF Transformers · VeRL · vLLM · Strands Agents

**AI Operations:** LLM fine-tuning · Multi-agent orchestration · Prompt engineering · RAG · Model Context Protocol

**ML Packages:** PyTorch · Scikit-learn · Pandas · SciPy · OpenCV

**Software Tools:** Git · Docker · SQL · MATLAB · LabVIEW · PuLP Solver

**Other:** ROS · Open 5G stack · Arduino

## Leadership & Activities

- Peer reviewer for NeurIPS, AAAI, ICLR, L4DC, and IEEE ComSoc
- Electrical Engineering Lead at UCR Unmanned Aerial Systems (July 2020 – June 2022): Led undergraduate engineering students in multiple projects to support the design and testing of fixed-wing UAVs